Greenwashing Moves to Where Enforcement Is Weakest: Pre-Registered Experiments in an Artificial Market

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Abstract

Greenwashing moves to where enforcement is weakest. We show this in pre-registered experiments in an agent-based market initialised on CSI 300 firms, where 1,000 investors price reported ESG scores and 200 firms choose between costly real improvement and cheap inflation that audits may detect. China's statutory fine leaves large firms inflating in two thirds of their decisions, and heavier fines add little, because inflation added to an undetected case is not fined again. When firms know the audit rule, auditing by market value moves inflation to small firms and auditing score jumps keeps it below the threshold; both do worse than random audits with the same budget. Letting firms choose their assurance provider moves inflation to the laxest one. At the doses tested, raising transparency improves true ESG more than quadrupling the fine, but the advantage falls from 0.2 to 0.003–0.012 standard deviations when true performance reverts to the mean.

AI disclosure: The simulation code, the analysis scripts and drafts of this paper were written with the assistance of Claude (Anthropic), an AI model, used through Claude Code. The author directed the research, reviewed all AI-assisted output and is responsible for the content.

Keywords: greenwashing; ESG disclosure; agent-based model; inspection game; impact investing; audit targeting; third-party assurance; pre-registration.
JEL: C63, G18, G23, K22, K42, M14, Q56.

*Fudan University. E-mail: kanzaler123@gmail.com. Preliminary; comments welcome. Code, configuration files, the pre-registrations and all evidence are at https://github.com/kanzaler123/financial-ABM. An in-browser version of the firm decision layer is at https://kanzaler123.github.io/financial-ABM/.

1 Introduction

This paper shows that greenwashing moves to where enforcement is weakest, and that this is largely why making inflated ESG reports easier to see through raises firms' true ESG performance more than raising fines does. We establish both in pre-registered experiments in an artificial stock market where 1,000 investors trade the shares of 200 firms initialised on CSI 300 constituents. Every period each firm chooses between costly real improvement, cheap inflation of its report that audits may detect, and doing nothing; impact investors pay for reported scores; and a regulator sets transparency, audits and fines. Neither instrument has been applied to ESG reports at scale and true performance is not observed, so we compare them in otherwise identical simulated markets.

The question is open in China. Exchange guidelines require several hundred large listed companies to publish sustainability reports from the 2025 reporting year but provide only disciplinary sanctions (e.g. Shanghai Stock Exchange, 2024), and it is not settled whether the Securities Law's fine for false disclosure, capped at CNY 10 million per case (Standing Committee of the National People's Congress, 2019, Art. 197), reaches ESG reports at all. That cap is about 0.01% of the median market value of a CSI 300 constituent. In the model, fixed fines of this kind miss in two places. Under the statutory fine, large firms inflate in 66% of their decisions and small firms in 6%. And moving from no fine to the statutory fine cuts the share of inflating decisions from 91% to 22%, but eight times that fine still leaves 12.5%, because a firm with an undetected case adds new inflation to it and faces no new fine. When every false report is fined separately, this plateau disappears.

Redirecting enforcement moves these gaps rather than closing them. At an unchanged expected number of audits, auditing in proportion to market value lowers the equal-weighted true level by 0.19 standard deviations (SD) relative to random audits, because greenwashing moves from large firms to the more numerous small firms, whose audit rate falls. Auditing firms whose reported scores jump lowers it by 0.14 SD, because firms keep new inflation below the jump threshold. Both rules find inflation more often than random audits. The loss appears to come mainly from firms' response to a known rule: with the rule hidden, neither differs significantly from random audits. Assurance providers show the same pattern. With strong liability and providers assigned at random, mandatory assurance raises the true level by 0.27 SD. When firms choose their provider, most of that gain is lost (−0.20 SD against assignment), and 96.9% of assured firms' inflating decisions are made by clients of the year's lowest-quality provider.

Transparency is harder to route around, because it lowers the payoff of every inflated report, whatever the firm's size, its history or who inspects it. Raising transparency from 0.1 to 0.6 raises the true level more than quadrupling the fixed fine, and both beat the baseline. With a common washing cost and lasting improvement, inflation stops altogether above a transparency of about 0.57, the threshold that an inspection-game benchmark of the firm's decision predicts (Section 4; Avenhaus et al., 2002). The doses are not matched on expected penalty or implementation cost, so this is a ranking at the doses tested, not a comparison at equal cost.

The advantage of transparency depends on real improvement lasting. The effect of transparency against the baseline is +0.04 SD without natural decay of the true level and +0.33 SD at 0.2 SD per year, and its advantage over the quadrupled fine is +0.205 SD at 0.2. When true levels instead revert to the cross-sectional mean, at two rates suggested by an unverified rating panel and used as sensitivity values, improvements fade while inflation persists, the transparency threshold rises above 0.6, and the advantage falls to +0.003 to +0.006 SD. A pre-registered test on new seeds that adds a fine for every false report keeps the ordering in all four settings, but the advantage stays at +0.004 to +0.012 SD. In our experiments the ordering of the two instruments is robust; the size of the advantage is not.

Two further results bound these conclusions. Every policy works only through investor preference:

without impact investors' taste for reported ESG, firms neither improve nor inflate and no policy has any effect, whereas with it the true level rises by 0.79 SD over fourteen years. And the market's response to policy is modest in the main setting. Replaying firms' decisions on the baseline run's prices and investor population shows that market feedback adds about 11% to the fixed-fine effect, and its net effect on the transparency policy cannot be distinguished from zero.

Design, seeds, metrics and primary questions were committed to a public repository before each confirmatory run: 59 scenarios with 30 paired seeds in the first experiment, 1,770 simulated markets of 3,500 trading days, and new seeds for the experiments on targeted audits and assurance providers and for the joint robustness test. Supplements designed after the first results are labelled exploratory, and each was written down before it was run. Every run is kept in the archive, failed ones included.

The results inherit the model's limits. Firms are myopic net-present-value maximisers. A tabular Q-learner trained on realised accounts failed a pre-set coverage test, and in a pre-registered redo it inflated under the information policy in 23.7% of decisions against 3.8% for an NPV firm, at a median loss of 0.042 basis points per decision, too small to learn from (Section 6.7); the main results say nothing about boundedly rational firms. The costs of transparency and audits are not modelled. And with the calibrated investor taste, a firm's financing rate responds to the signal investors see by about 2.2 basis points per SD, an order of magnitude below the empirical cost-of-equity association of about 28 basis points per SD that the calibration targets (Zhao et al., 2024).

The paper contributes to three literatures. Theories of greenwashing treat it as selective disclosure that an activist may audit (Lyon and Maxwell, 2011); we let firms choose how much to misreport under audits and fines, and show how the booking of fines and the visibility of audit rules decide where they do it. The closest agent-based work (Zhao et al., 2025, 2026) derives firms' equilibrium upgrade strategies under strict and lax disclosure; there, greenwashing arises passively from pooled disclosure categories and there are no audits or fines, whereas our firms misreport actively, differ in size and finance themselves in a market whose prices and investor population respond to policy. Equilibrium models of sustainable investing show how investors' taste lowers green firms' cost of capital (Heinkel et al., 2001; Pástor et al., 2021; Pedersen et al., 2021); we take that premium as the reward for both real improvement and greenwashing and ask which instrument tilts the balance between them. We also bring pre-analysis plans (Olken, 2015) to agent-based policy analysis. Section 2 discusses related work in more detail.

2 Related literature

Greenwashing and ESG disclosure. Lyon and Maxwell (2011) model greenwash as selective disclosure by a firm that an activist may audit, and Delmas and Burbano (2011) and Lyon and Montgomery (2015) review its drivers and forms. Empirically, firms both exaggerate and understate their sustainability performance (Kim and Lyon, 2015), and selective disclosure is less common where firms face more scrutiny (Marquis et al., 2016). On disclosure mandates, Christensen et al. (2021) review the economics of mandatory sustainability reporting; Chen et al. (2018) find that China's 2008 mandate lowered the profitability of affected firms and reduced pollution in the cities most exposed to them; Fiechter et al. (2022) find that the EU directive raised affected firms' CSR activity. ESG ratings of the same firm diverge widely across providers (Berg et al., 2022), one reason we treat the reported score as a proxy rather than the true attribute. Our information instrument is a stylised version of these mechanisms; unlike reduced-form studies, we can switch it on in otherwise identical markets and compare it with fines.

Enforcement. The deterrence logic of fines goes back to Becker (1968); inspection games formalise the interaction of an inspector who audits with some probability and an inspectee who may violate

(Avenhaus et al., 2002). We use the inspectee's side as an analytical benchmark for the firm layer. The audit probability in our model is an assumption; as background, Dyck et al. (2024) estimate that only about a third of corporate fraud is detected.

Sustainable investing and the cost of capital. When some investors exclude polluting firms, those firms' cost of capital rises and they may reform (Heinkel et al., 2001); in equilibrium, investors' taste for green holdings lowers the expected returns of green assets (Pástor et al., 2021; Pedersen et al., 2021). Empirical estimates of the financing effect range from 7–18 basis points on US bank loans (Goss and Roberts, 2011) to about 0.19 percentage points of the cost of equity per rating notch for Chinese listed firms (Zhao et al., 2024), which we use as the calibration target. In our model the premium is both the reward for real improvement and the reward for greenwashing, and the share of investors who pay it evolves.

Agent-based models of green disclosure. The closest work is by Zhao et al. (2025) and Zhao et al. (2026). They simulate firms at three green levels that compete for a fixed pool of ESG fund capital allocated by investors' valuation of each level, and they use empirical game-theoretic analysis to find equilibrium upgrade strategies under strict and lax disclosure and under a two-stage policy that switches from lax to strict. The journal version extends it to a dual-market model of the EU and China. In their model a firm decides only whether to upgrade. Greenwashing is passive: lax disclosure pools low- and high-green firms at their average valuation, and no firm chooses to misreport. There are no audits or fines, and firms differ by green level rather than size. Our paper differs in four respects. Firms actively choose how much to inflate a report, at a cost and under a risk of detection. Enforcement has an explicit audit-and-penalty structure, in which fines are booked per case, fixed and proportional fines are compared, and the statutory cap is calibrated. Firm size matters because a lump-sum fine shrinks relative to value as firms grow, which produces the size divide of Section 6.5. And financing gains come from a market whose prices and investor population respond to policy, so we can measure how much that feedback contributes (Section 6.6). Their method answers a question our simulation does not: it builds an empirical payoff table from simulations and applies α-Rank, which ranks strategy profiles by their weight in the stationary distribution of an evolutionary dynamic, so it says which profiles a population of firms would persist in.

Agent-based finance and research practice. Artificial stock markets with heterogeneous, adaptive agents go back to Arthur et al. (1997); LeBaron (2006) surveys the field and Farmer and Foley (2009) argue for its use in policy analysis. We parameterise demand elasticities in the range of the demand-system literature (Koijen and Yogo, 2019) and describe the model following the logic of the ODD protocol (Grimm et al., 2020). Agent-based studies leave many choices to the researcher; following the case for pre-analysis plans in economics (Olken, 2015), we fixed design, seeds and inference before the confirmatory run and keep failed runs in the record.

3 The model

3.1 Overview and timing

A simulation runs for 3,500 trading days (252 per year, about 14 years). Day 0 initialises the market; days 1–1200 are a pre-policy period under the baseline regime; a policy, if any, switches on at day 1201 and stays in place. Outcomes are measured over the policy window, days 1200–3500. The agents are 1,000 investors of two types, 200 firms in three size tiers (the largest 10%, the next 30% and the remaining 60%

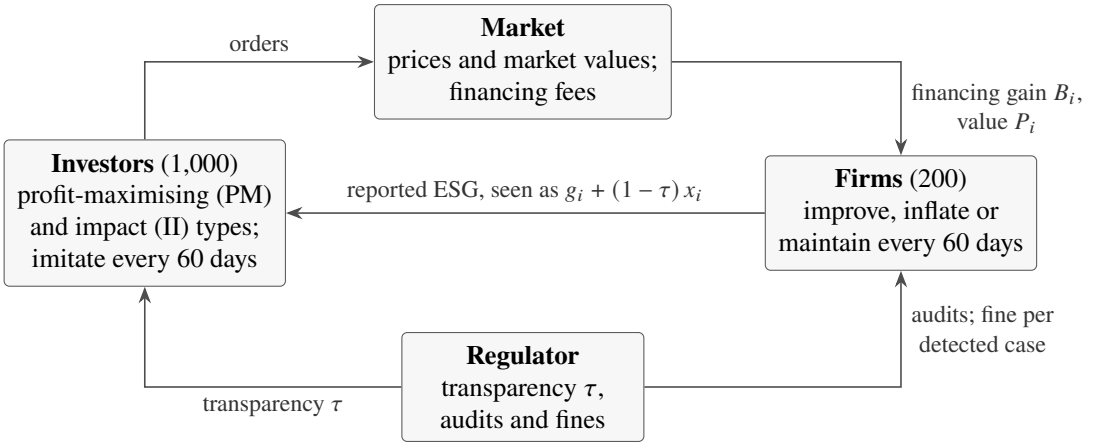


Figure 1: Structure of the model. A firm's true level is g_i and its undetected inflation x_i; investors see the inflation only in part, $1-\tau$. Firms affect each other and are affected by investors only through the financing gain B_i and their market value P_i.

by initial value), market makers with an aggregate noise-trader account, and a regulator. Figure 1 shows how they interact. Table 1 lists the main parameters with their sources.

3.2 Firms and the disclosure state

Each firm i has a latent true level g_i, measured in units of its initial cross-sectional standard deviation (SD), and undetected inflation $x_i \geq 0$; its report is $g_i + x_i$. Investors observe the signal

$$s_i = \min\{\, g_i + (1-\tau)\, x_i,\ C \,\}, \tag{1}$$

where $\tau \in [0,1]$ is transparency, the share of inflation that investors see through, and C is the latent value of a perfect score of 100 points. Reports cannot exceed a perfect score, so inflation is capped at $\max\{C - g_i, 0\}$. Without effort, the true level decays by δ SD per year (main setting $\delta = 0.2$).

Every 60 trading days each firm takes one of three actions. *Improve* by Δ costs $c_i\Delta^2$ and raises g_i after an implementation delay of 120 days. *Inflate* by Δ costs $w\, c_i\Delta^2$ and enters the published report after 20 days. *Maintain* costs nothing. Changes are capped at 0.25 SD per period. Improvement beyond the report's remaining room replaces inflation, so only its transparent share is newly visible. The marginal cost scale is $c_i = \bar{c}\, m_{k(i)}\, \varepsilon_i\, e^{\eta g_i}$, with $\bar{c} = 0.05$, size multipliers $m = (0.6, 1, 1.6)$ for large, mid and small firms, lognormal dispersion ε_i with log-SD 0.5, and $\eta = 1.5$, so improvement becomes dearer as the level rises. The size multipliers assume that large firms improve more cheaply; an equal-cost control removes them. The washing cost ratio $w = 0.2$ has no empirical basis and is varied in the experiment. Costs, fines and subsidies change a firm's intrinsic value and reach its price through the market makers on the next day.

3.3 Investors and prices

Investors differ in wealth (lognormal), risk aversion γ_j and decision calendar (median 20 days). On a decision day an investor sets its equity share to the Merton ratio $(E[R_m] - r_f)/(\gamma_j \sigma_m^2)$, clipped to $[0,1]$, and allocates equity across firms by a softmax tilt away from market weights,

$$w_{ji} \propto P_i \exp(\kappa_j U_{ji}), \qquad U_{ji} = E[R_i] - \gamma_j \operatorname{cov}(r_i, r_m) + \theta_j\, s_i, \tag{2}$$

moving half-way from its current to its target holdings. Profit-maximising investors (PM) have $\theta_j = 0$. Impact investors (II) have a lognormal taste θ_j, measured as an annual-return equivalent per SD of the signal, with median $\bar{\theta}$ and log-SD 0.5. With equal utilities every investor holds the market portfolio, which keeps the rule consistent with a fixed supply of shares.

Each firm's shares are quoted by a market maker who passes through news about intrinsic value and moves the price with order flow. Price impact has a permanent and a transient part, depth thins when volatility rises, and an aggregate noise-trader account adds correlated flow. Seven market parameters were fitted by the method of simulated moments to return moments of US industry portfolios; cash and shares are conserved every day. The market layer was frozen before any policy experiment, and its known defects are reported in Section 7.

3.4 Investor evolution

Every 60 days, 10% of investors review their type. Each reviewer samples one other investor uniformly; if the partner is of the other type, the reviewer switches with probability $k \cdot \max\{0, \bar{G}_{\text{other}} - \bar{G}_{\text{own}}\}$, where $\bar{G}$ is the mean wealth growth of each type over the window and $k = 100$. This is the proportional imitation rule of Schlag (1998). The expected change in the impact-investor share s is $m\,k\,s(1-s)\,E[\bar{G}_{II} - \bar{G}_{PM}]$ with review share m, a discrete replicator dynamic (Taylor and Jonker, 1978), so the share does not drift when the two types perform equally. An earlier implementation omitted the partner draw and drifted towards one half even at zero taste. With the corrected rule, the change in the share over 14 years at zero taste has 95% intervals that include zero at initial shares of 0.2, 0.5 and 0.8 (20 seeds each), and a taste near the calibrated value raises it by 2.3 percentage points [1.4, 3.1] from an initial share of 0.2: the market's own drive towards impact investing is small.

3.5 Financing gains and the firm's decision rule

Each firm has a fixed external financing exposure E_i, half of its initial market value, whose fee rate $r_f + 0.05\,E_i/(E_i + P_i^{\text{lag}})$ is repriced every 60 days. A firm estimates the local permanent response of its price to its own signal, holding everything else fixed, and converts it into B_i, the annual financing saving per SD of signal as a share of current market value. Discounting at 5% a year, the calendar of delays, repricing dates and audits gives three factors. The signal annuities of real improvement, A_r, and of inflation, A_w (which lasts only until an audit), count financing savings over a five-year horizon. The expected discounted value D of paying one unit when new inflation is detected has no horizon: a false report remains a liability until an audit finds it, so D sums the audit probability over all future audit dates. With subsidy σ per SD, proportional fine p per SD of detected inflation, fixed fine F and market value P, the values of the three actions are

$$V_I = \frac{(BA_r + \sigma)^2}{4c}, \tag{3}$$

$$V_W = \frac{\max\{B(1-\tau)A_w - pD,\, 0\}^2}{4wc} - \pi\frac{F}{P}D, \tag{4}$$

$$V_M = 0,$$

with optimal sizes $(BA_r + \sigma)/2c$ and $(B(1-\tau)A_w - pD)/2wc$ before caps. The term π is the probability that new inflation opens a new case, which is zero for a firm that already carries undetected inflation (see below). The firm takes the action with the highest value and maintains on ties. The rule is myopic: firms re-optimise every period and do not anticipate their own future choices.

3.6 The regulator

In each 60-day period every firm is audited with probability 1%. An audit that finds inflation opens a case: the firm pays the fixed fine F plus p times its inflation, and the inflation is removed. Inflation added while earlier inflation is still undetected joins the same case and adds no fixed fine. We call this *per-case booking*; the alternative, a separate fine for every false report, is tested in an exploratory supplement (Appendix A.2). In the main setting $p = 0$ and F equals CNY 10 million, the statutory maximum, which is 4.56×10^{-5} of the average firm's initial value. Baseline transparency is $\tau = 0.1$. The second experiment (Section 8) replaces random audits with targeted rules.

3.7 Initialisation and calibration

The market is initialised on the CSI 300 constituents of 9 October 2026, matched 300 of 300 on Huazheng composite ESG scores, half-year book equity and market value (Table 1). Scores are mapped to the latent standard-normal level by their empirical quantiles, so the scale saturates at a perfect score, and the latent level is linked to log value by a Gaussian copula with $\rho = 0.177$, converted from the sample Spearman correlation of 0.169. The dispersion of log intrinsic value, 0.915, is the cross-sectional SD of log market value.

One parameter was calibrated to a target. We set the median taste $\bar{\theta} = 0.0724$ so that, before any policy, the cross-sectional slope of dividend yield on the signal is about −28 basis points per SD (validation median −28.25). The target combines the estimate of Zhao et al. (2024), 0.19 percentage points of the cost of equity per Huazheng rating notch, with the 1.45 notches per SD in our sample; it is a cross-sectional association, not a causal effect. The audit probability of 1% per period is an assumption. It implies a discounted value of about 0.455 for paying one unit on detection; this is a present-value factor, not a detection rate, and is not comparable with the estimate of Dyck et al. (2024) that about a third of corporate fraud is detected, which we cite only as evidence that detection is far from certain. The remaining firm-layer parameters — natural decay, the washing cost ratio, the cost function, delays and the financing contract — are also assumptions, and the experiment varies the ones that matter most.

4 Analytical benchmark

The firm's decision rule is the inspectee's best response in an inspection game with given audit probability, fines and transparency. Two propositions summarise it; both describe the rule in equations (3)–(4) for an interior solution.

Proposition 1 (Inflating under a fixed fine) *Let $\sigma = 0$, $p = 0$, $\pi = 1$ and $\tau < \tau^*$ (Proposition 2). A firm inflates if and only if*

$$B^2\Big[\frac{(1-\tau)^2 A_w^2}{w} - A_r^2\Big] > 4\,c\,\frac{F}{P}\,D,$$

that is, if and only if its market value exceeds $P^* = 4cFD\,/\,\big(B^2[(1-\tau)^2 A_w^2/w - A_r^2]\big)$.

Proof. With $B > 0$, $V_I > 0$, so inflating must beat improving: $B^2(1-\tau)^2 A_w^2/4wc - FD/P > B^2 A_r^2/4c$. □

Three consequences follow. The fixed fine decides *whether* a firm inflates, not *how much*, unlike a proportional fine, which lowers the net gain per SD. Because F/P falls with size while B is roughly size-independent and large firms improve more cheaply, large firms cross P^* first; under a proportional fine the condition becomes $B[(1-\tau)A_w - \sqrt{w}A_r] > pD$, in which size enters only through B. And for a

Table 1: Main parameters

Parameter	Main setting	Source or status
Scale	1,000 investors; 200 firms; 3,500 days	Design; policy switch on day 1201
Initial ESG distribution	Empirical quantiles of Huazheng scores; perfect score at $C = 2.576$	Fitted, CSI 300 constituents (300/300)
Size–ESG dependence	Gaussian copula, $\rho = 0.177$	Fitted, Spearman 0.169
Dispersion of log value	0.915	Fitted, SD of log market value
Impact-investor taste $\bar{\theta}$	0.0724 (log-SD 0.5)	Calibrated: −28 bp/SD yield slope
Initial impact-investor share	0.2 (each investor II w.p. 0.2)	Assumption; 0–0.8 in phase diagram
Imitation	Every 60 days, 10% review, $k = 100$	Assumption; passes zero-taste drift test
Fixed fine F	CNY 10 million per detected case	Statutory maximum, Securities Law Art. 197
Audit probability	1% per 60-day period	Assumption
Transparency τ	0.1	Assumption
Natural decay δ	0.2 SD per year	Assumption; 0–0.2 tested
Washing cost ratio w	0.2	Assumption; 0.05–0.4 tested
Cost of improvement	$\bar{c} = 0.05$; size multipliers (0.6, 1, 1.6); $\eta = 1.5$	Assumption; equal-cost control
Delays and caps	120 days (improve), 20 days (report); 0.25 SD per period	Assumption
Financing contract	Exposure 0.5, spread 0.05, 5-year horizon, 5% discount	Assumption
Market microstructure	Seven parameters	Simulated moments, US industry returns

Notes: SD is the initial cross-sectional standard deviation of the latent true level. "Fitted" parameters reproduce a feature of the CSI 300 sample; "calibrated" means matched to an empirical target before any policy result was seen; assumptions are not estimated.

firm with an open, undetected case ($\pi = 0$) the condition reduces to $\tau < \tau^*$ and no longer depends on F: a higher fine can stop firms from starting to inflate but not from continuing. At an audit probability of 1% per period, an undetected case lasts about 100 periods on average.

Proposition 2 (Transparency threshold) *Let $\sigma = 0$ and $p = 0$. Firm i inflates if and only if $\tau < \tau_i^*$, where*

$$\tau_i^* = 1 - \frac{\sqrt{w}}{A_w}\sqrt{A_r^2 + \frac{4\, c_i\, \pi_i\, (F/P_i)\, D}{B_i^2}},$$

and $\tau_i^* \le \tau^* \equiv 1 - \sqrt{w}\, A_r / A_w$, with equality if and only if $\pi_i F = 0$.

Proof. Solve the inequality of Proposition 1 for $1 - \tau$. □

Without a fine, every firm has the same threshold τ^*, which depends only on w and the timing that determines A_w/A_r, not on costs, size or financing gains. With the default timing $A_w/A_r \approx 1.03$, so $\tau^*(0.2) \approx 0.57$. A fixed fine spreads the thresholds below τ^*, and firms that already inflate keep the threshold τ^*.

The propositions hold the financing gain B fixed and ignore other firms. In the simulation, B_i and P_i move with prices and with the investor population, firms carry different states and open cases, and policies change all of these. The experiment therefore asks two things the propositions cannot answer: how large the policy effects are once financing gains come from a market, and how much the feedback from policy to B and P changes them. The threshold prediction itself is close to a property of the decision rule: above $\tau^*(w)$ no firm can find inflation worthwhile whatever B is. Its confirmation in the simulation (Section 6.4) is therefore a check of the implementation under market feedback rather than independent evidence.

Table 2: Scenario matrix of the confirmatory experiment

Block	Scenarios	Count
A Main comparison	B, I ($\tau = 0.6$), F (fixed fine 4×), IF; the same four with taste set to zero ("market off")	8
B Doses	$\tau \in \{0.3, 0.5, 0.7, 0.8\}$; fixed fine 0×, 2×, 8×; proportional fine 1×, 2×, 4×	10
C Size identification	Equal cost multipliers (1, 1, 1) × B, I, F, IF	4
D Null controls	No impact investors; no taste and no policy from day 0	2
E Decay control	No natural decay × B, I, F, IF	4
F Initialisation controls	Size–ESG independence; book-equity dispersion (1.298) × B, I, F	6
G Phase diagram 1	Taste $\{0, 0.036, 0.072, 0.145\}$ × initial II share $\{0, 0.2, 0.5, 0.8\}$	13
H Phase diagram 2	Washing cost $w \in \{0.05, 0.1, 0.2, 0.4\} \times \tau \in \{0.1, 0.3, 0.6, 0.8\}$	12

Notes: Each scenario is run with the same 30 seeds. Phase-diagram cells that coincide with scenarios in other blocks reuse them, hence 13 and 12 new scenarios. Within a market setting, all scenarios share the same history up to day 1200 bit for bit.

5 Experimental design

Pre-registration. The design was written into the project's research contract (Section 20) and committed (`2b08f65`) before the confirmatory run. A development run on separate seeds checked only the pipeline: completion, a null control with no actions, bit-identical pre-switch histories and metric windows. The confirmatory configuration was then frozen and identified by the hash of its contents (`90ee8606...5d57`). The confirmatory seeds, 41001–41030, appeared in no seed field of the 6,039 configuration and archive files from earlier work.

Scenarios. Table 2 lists the 59 scenarios. The four main policies are the baseline (B), information (I: transparency raised from 0.1 to 0.6), enforcement (F: fixed fine raised to four times the statutory maximum, i.e. CNY 40 million, as if the law were amended), and both (IF). Doses were fixed in advance and were not matched by expected penalty or cost. The proportional fine uses an *equal-amount reference*: for a firm of average value, 1 SD of detected inflation costs CNY 10 million, so the two fines differ only in whether the amount grows with size and with the amount of inflation.

Outcomes and questions. The primary metric is the change in the equal-weighted mean true level from day 1200 to day 3500, in initial SD. Four primary questions were tested two-sided with the Holm (1979) correction:

P1 market preference: B minus market-off B, over the full period (days 0–3500);

P2 instrument comparison: I minus F;

P3 interaction: (IF − B) − (market-off IF − market-off B);

P4 size mechanism: B minus equal-cost B, in the large-minus-small difference of the change.

The other 36 contrasts are reported with unadjusted intervals and make no confirmatory claim. Because all scenarios in a market setting share their history up to the switch, contrasts are paired by seed, and intervals are paired 95% t intervals over the 30 seeds.

Phase classification. A world is classified by its terminal-year change in the true level net of the decay it would have had without action (G) and its terminal-year inflation (W), with threshold 0.25 SD: real improvement ($G \geq 0.25$, $W < 0.25$), greenwashing trap ($G < 0.25$, $W \geq 0.25$), low improvement and

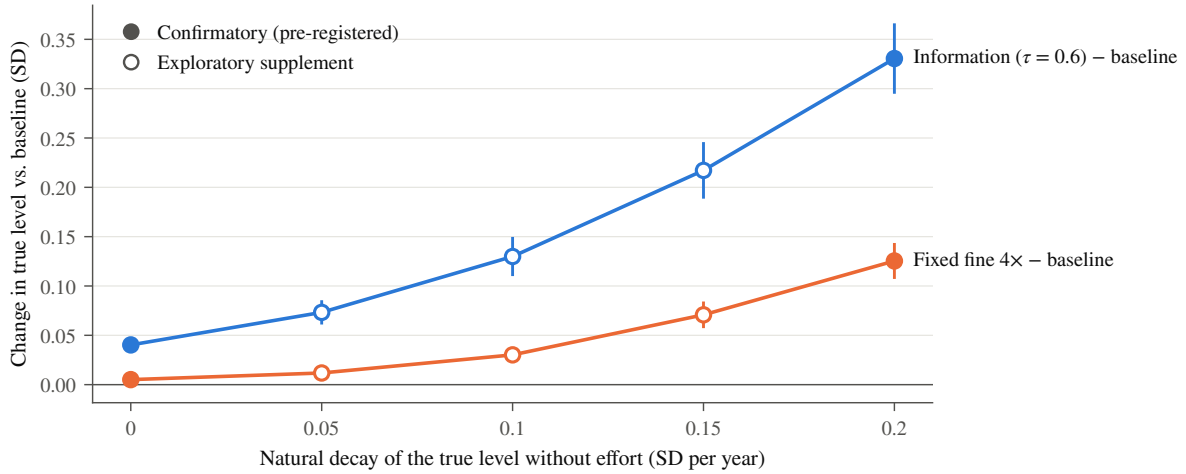


Figure 2: Policy effects on the true level over the policy window (days 1200–3500) at five uniform rates of natural decay. Points are means over 30 paired seeds; bars are paired 95% t intervals. Filled points are confirmatory; hollow points come from the exploratory decay supplement.

low inflation, or mixed. A cell is labelled with a state only if at least 80% of its seeds share it. All nine combinations of thresholds in $\{0.125, 0.25, 0.5\}$ are in the archive.

Failures. Every run is kept. Of 1,770 runs, all completed and 37 triggered the numerical price guard, a mechanical validation gate; they remain in the statistics, and excluding them leaves P2 at +0.207 and P3 at +0.333.

Exploratory supplements. After the confirmatory results we ran five supplements, each written into the contract and committed before its own implementation and runs: three intermediate decay rates (360 runs); a counterfactual that removes market feedback (360 reruns and replays); the same decomposition in stronger markets (270 runs); a split of the feedback into two channels (240 runs); and three threats to the main claims, namely mean-reverting true levels, a fine for every false report and firm-specific washing costs (600 runs). New options default to the original behaviour, and rerunning stage-F scenarios with them off reproduces the archive bit for bit. They use the same seeds and carry no confirmatory weight.

6 Results

6.1 How to read magnitudes: natural decay

Natural decay matters because it makes room for inflation: a firm that does nothing sees its score slide and can restore it cheaply by inflating. Figure 2 shows the two policy effects at the five tested decay rates. Without decay, only 4.7% of baseline decisions inflate and the effect of the information policy is +0.040 SD; at 0.2 SD per year, 22.3% inflate and the effect is +0.331 SD. The fine's effect grows from +0.005 to +0.125 SD and the gap between the two instruments from +0.035 to +0.205 SD. Every interval excludes zero, and the ordering I > F > B holds at all five rates: at the two pre-registered end points and at the three intermediate rates of the exploratory supplement. The growth is convex: at 0.1 SD per year the information effect is 39% of its value at 0.2. Unless decay is close to 0.2, effects are closer to the lower end of the range.

We therefore report magnitudes as the range from no decay to 0.2 SD per year. Results available

Table 3: Primary questions

Question	Contrast	Mean (SD)	95% interval	Holm p
P1 Market preference	B − market-off B, days 0–3500	+0.795	[+0.743, +0.847]	3×10^{-23}
P2 Instrument comparison	I − F	+0.205	[+0.176, +0.235]	1×10^{-14}
P3 Interaction	(IF − B) − (off IF − off B)	+0.331	[+0.295, +0.366]	1×10^{-17}
P4 Size mechanism	B − equal-cost B, large − small	+0.229	[+0.205, +0.252]	6×10^{-18}

Notes: Main setting (natural decay 0.2 SD per year), 30 paired seeds. Contrasts P2–P4 use the policy window, days 1200–3500. One run in each of I and IF triggered the price guard; excluding them gives P2 = +0.207 and P3 = +0.333.

only for the main setting — doses, proportional fines, size and the phase diagrams — hold under the assumption of 0.2 SD per year and are cited with that condition. The three main contrasts grow with decay, but we have not tested how these other results vary with it, so we do not claim they are upper bounds. Absolute levels are not interpretable: under the assumed decay every scenario declines (B by −1.196 SD), and with market preference switched off the decline, −1.825 SD, is exactly the decay accumulated over 2,300 days. Only differences between scenarios carry information.

Uniform decay may also be the wrong form. In an unverified panel of Chinese ESG ratings, ratings show no trend and revert towards the middle. When true levels instead revert to the cross-sectional mean at two data-motivated sensitivity rates, the ordering I > F > B survives, but the advantage of information over the quadrupled fine falls to +0.003 to +0.006 SD: reversion erodes real improvement while inflation is removed only by audits, and transparency of 0.6 no longer stops inflation (Appendix A.1).

6.2 Investor preference is necessary

Impact investors' taste raises the true level over the full period by +0.795 SD [+0.743, +0.847] (P1, Holm $p = 3 \times 10^{-23}$; Table 3). With taste switched off, firms take no action under any policy: improvement earns nothing and inflation is not needed to hold the signal, so the four policies produce identical worlds. The interaction P3, +0.331 [+0.295, +0.366], therefore equals the effect of IF with the market on. It is mechanical rather than a finding about how two forces combine: in this model, policy acts only through the preference of investors.

Taste with a positive impact-investor share always produced a mixed state under the baseline policy: in every such cell of phase diagram 1 the dominant state was mixed, with decay-adjusted improvement and inflation both above 0.25 SD, and no cell reached improvement without inflation. The investor population itself moves slowly. Over 14 years the impact-investor share rises by only a few percentage points (Section 3), so the transition depends on the preference that investors already have rather than on the market recruiting new impact investors.

6.3 Information versus enforcement

Table 4 summarises the four main policies. Information beats enforcement by +0.205 SD [+0.176, +0.235] (P2), and both beat the baseline. Across the decay range, the gap is +0.035 to +0.205 SD. The information policy stops inflation altogether and, because baseline spending goes largely into inflation, firms spend *less* on ESG under it (2.57 against 3.77 basis points of initial market value). Fines collected are tiny, 0.03–0.14 basis points of initial value: fines work through deterrence, not through the money they move.

Figure 3 places every tested dose of each instrument in one plane: the share of decisions that inflate against the change in the true level relative to the baseline. Transparency of 0.3 and 0.5 raises the true level by +0.181 and +0.259 SD; from 0.6 on, inflation stops and further transparency adds almost nothing

Table 4: Main policies, policy window

Scenario	True change (SD)	Inflating decisions all (large / mid / small)	Mean inflation (SD)	Spending (bp)	Fines (bp)
B Baseline	−1.196	22.3% (66 / 41 / 6)	0.95	3.77	0.05
I Information, $\tau = 0.6$	−0.865	0	0.43	2.57	0.03
F Fixed fine 4×	−1.070	13.3% (54 / 22 / 2)	0.77	2.98	0.14
IF Both	−0.865	0	0.43	2.57	0.13
Market off (all four)	−1.825	0	0	0	0

Notes: Means over 30 seeds; main setting. True change is negative in every scenario because of the assumed decay of 0.2 SD per year; only differences between rows are interpretable. Spending is firms' resource cost of improvement and inflation, and fines are fines paid, both in basis points of initial total market value.

(+0.331 to +0.340). Fixed fines of 2, 4 and 8 times the statutory maximum raise it by +0.088, +0.125 and +0.143, and the share of inflating decisions falls to 15.5%, 13.3% and 12.5%. Within the tested range, no fine dose pushes inflation below 12.5%, while transparency at or above 0.6 eliminates it. This statement does not depend on how the two doses are matched, but it does depend on per-case booking: once a firm carries an undetected case, new inflation joins it and no fixed fine is added, so higher fines can only stop firms from opening new cases (Proposition 1). The plateau is an artefact of that accounting. When every false report is fined separately, eight times the statutory fine leaves 0.8% of decisions inflating, and the gap between information and the quadrupled fine shrinks from +0.205 to +0.058 SD, still positive (Appendix A.2). Which booking is right is a legal question.

The comparison is a comparison at fixed doses. It is not a comparison at equal expected penalty — transparency imposes no penalty — nor at equal social cost, because the costs of standards, assurance and audits are not modelled.

6.4 The transparency threshold

Figure 4 shows the share of inflating decisions on the grid of washing cost ratios and transparency levels, with the threshold $\tau^*(w)$ predicted by Proposition 2. In all 16 cells the prediction holds: no decision inflates once $\tau \geq \tau^*(w)$, and some do below it. Below the threshold the share declines gradually with transparency, because fines spread firm thresholds. Inflation already in reports does not vanish at the switch: at $w = 0.2$, terminal-year inflation is 1.33 SD at the baseline transparency of 0.1 and 0.35 SD at 0.6, not zero, since only audits, at 1% per period, or real improvement remove old inflation. Phase diagram 2 has a region of real improvement only where washing is expensive ($w = 0.4$) and transparency is at least 0.6: in each of these two cells, 27 of the 29 classified seeds are real improvement (one seed per cell failed core validation and is not classified).

Three qualifications matter for how this result is used. First, as Section 4 explains, the upper bound of the threshold depends only on w and timing, which the market cannot move; the 16 of 16 match confirms that the simulation implements the rule under feedback, not that real firms behave this way. Second, the threshold is sharp because every firm has the same washing cost ratio. If the ratio differs across firms, the aggregate response is governed by those that inflate most cheaply: in the cells with $w = 0.05$, 19% of decisions still inflate at $\tau = 0.6$, and inflation stops only near 0.78. Third, the threshold assumes that real improvement lasts. Under mean reversion it rises to about 0.8 (Appendix A.1). With firm-specific washing costs, inflation declines firm by firm instead of stopping at a common threshold, and the policy comparison barely moves (Appendix A.3).

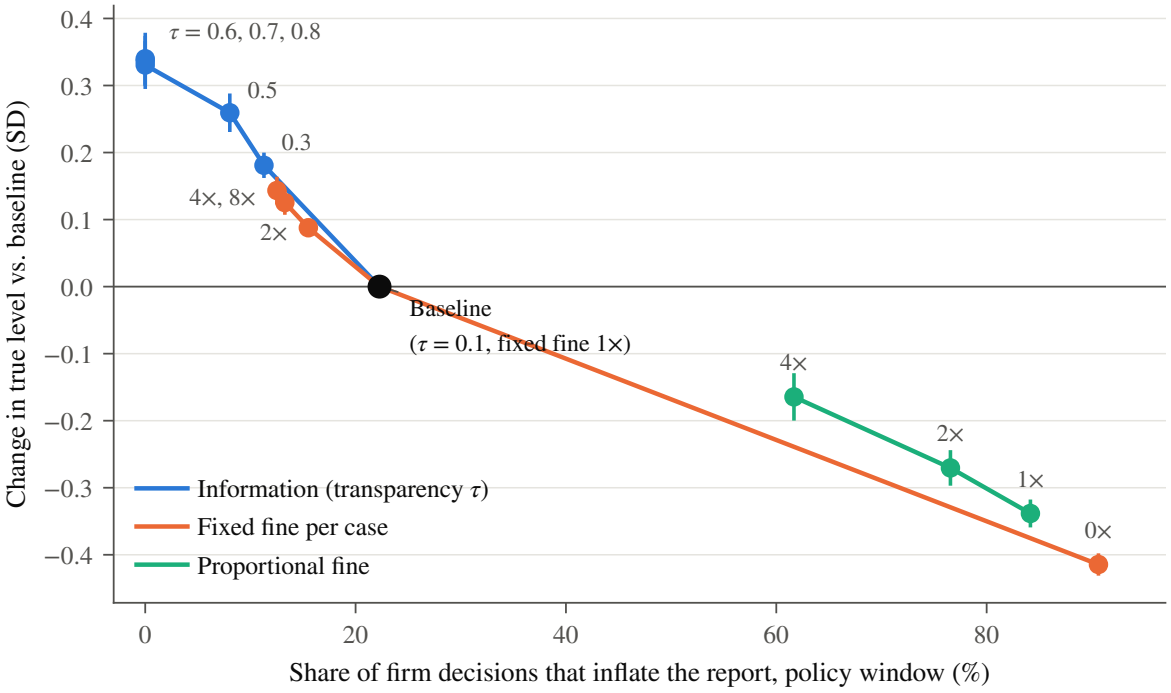


Figure 3: Every tested dose of the three instruments: share of firm decisions in the policy window that inflate the report, against the change in the true level relative to the baseline (paired 95% intervals). The baseline is transparency 0.1 with the statutory fixed fine. Main setting.

6.5 Fine design and the size divide

The size statistic in this section is the large-minus-small difference in the *change* of the true level over the policy window, days 1200–3500; it says which firms improve more during the window, not how firms rank at its end. Under the baseline, the true level of large firms changes by 0.33 SD less than that of small firms, because large firms inflate in 66% of their decisions and small firms in 6%. The fine is the reason: CNY 10 million is a negligible share of a large firm's value, so large firms cross the value threshold P^* of Proposition 1 first. The cost advantage assumed for large firms works the other way. Removing it widens the difference from −0.33 to −0.56 SD (P4: +0.229 [+0.205, +0.252]), so the difference comes from inflation, not from costs. Under the information policy the sign changes: large firms' true level changes by 0.38 SD *more* than small firms' over the window.

Figure 5 contrasts the two fine designs. Raising the fixed fine eightfold lowers inflation in every tier but still leaves large firms inflating in 48% of their decisions (66% at the statutory fine). A proportional fine reverses the size pattern, as Proposition 1 predicts: large firms now inflate least (72% against 88% for small firms at 1×). But at the equal-amount reference it is much weaker overall. It leaves 84% of decisions inflating and lowers the true change relative to the fixed fine of the same multiple by −0.338 SD [−0.359, −0.318] at 1×, −0.358 at 2× and −0.290 at 4×. The reason is the reference point: per-period inflation is far below 1 SD, so a fine of CNY 10 million per SD of inflation is much lighter per decision than a lump sum of the same amount. The result shows how the two designs differ at this reference; it does not show that either is stronger in general.

6.6 What the market adds

A firm's decision depends on other firms and on investors only through its financing gain B_i and its market value P_i, and audit draws come from a separate random stream shared across scenarios. Given

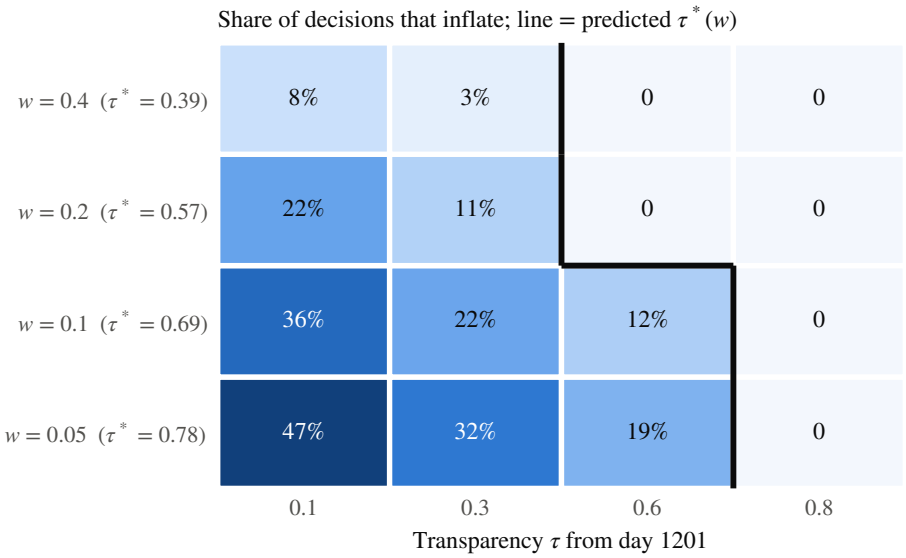


Figure 4: Share of firm decisions in the policy window that inflate the report, by washing cost ratio w and transparency τ (means over 30 seeds). The black line marks the threshold $\tau^*(w)$ of Proposition 2; the prediction is that no firm inflates to the right of it. Main setting.

Table 5: Contribution of market feedback to policy effects (true change, SD)

	Main setting		Taste doubled (0.145), II share 0.2	
Contrast	Feedback	Share of effect	Feedback	Share of effect
I − B	−0.008	not identified	−0.024	−4%
F − B	+0.014	11%	+0.049	31%
I − F	−0.021	−10%	−0.073	−18%

Notes: Feedback is the full-model effect minus the effect when the firm layer is replayed on the baseline run's financing gains and market values. Exploratory supplements; 30 paired seeds.

the paths of B and P and the audit draws, the firm layer can therefore be replayed without the market. Replaying each run on its own paths reproduces it bit for bit (360 of 360). Replaying a policy scenario on the *baseline* run's paths gives the policy effect without feedback, and the difference from the full model is the contribution of feedback from policy to prices, financing gains and the investor population.

In the main setting this feedback is small and its sign differs by contrast (Table 5). Its net effect on the information policy cannot be distinguished from zero (−0.008 SD [−0.022, +0.007]). It strengthens the fixed-fine effect by 11% of that effect (+0.014 SD) and narrows the gap between the instruments by 10% (−0.021 SD). The equal-cost and no-decay settings give shares of the same order (10–13% where identifiable). Where investor taste is doubled, feedback strengthens the fixed-fine effect by 31% and narrows the gap by 18%. Freezing the investor population splits the feedback into two channels. In the main setting, investor evolution weakens the information policy (−0.040 SD [−0.054, −0.026], −12% of its effect), while price and demand responses strengthen it (+0.032 [+0.028, +0.037], +9.8%); the two nearly cancel, and the information policy leaves the impact-investor share 2.0 percentage points lower at the end. The ordering I > F > B holds in 30 of 30 seeds with and without feedback.

The market matters more through the *level* of financing gains it produces than through feedback. Replaying the firm layer on a single snapshot of B and P taken at day 1200, as the browser version of the model does, reproduces the five qualitative patterns we checked but overstates several magnitudes

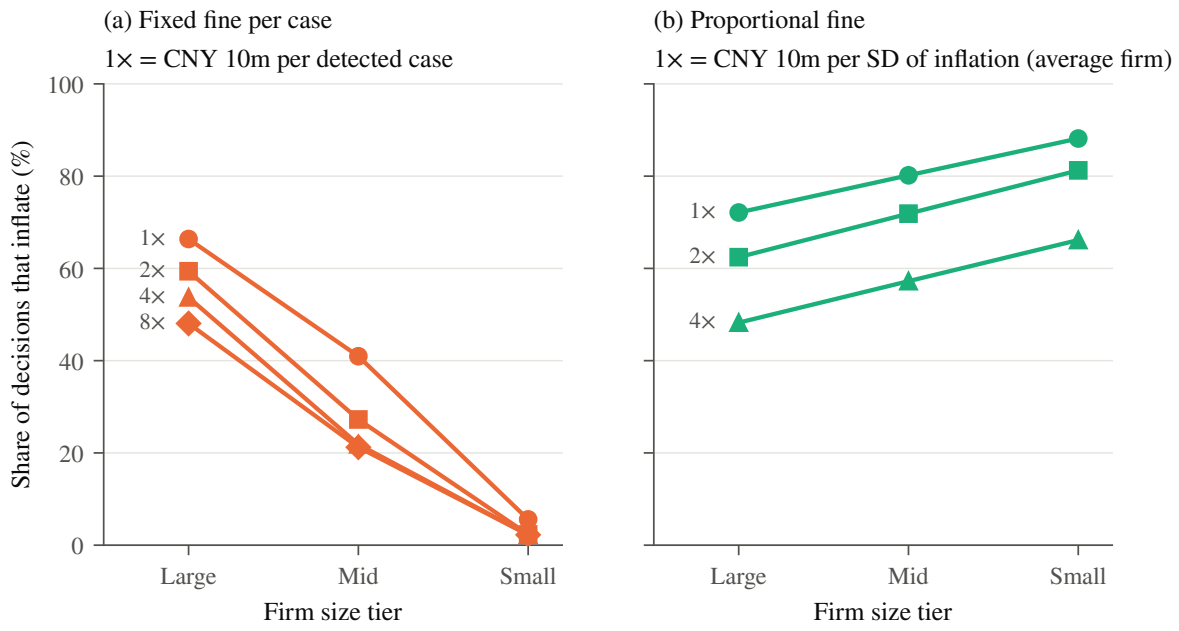


Figure 5: Share of decisions that inflate by firm size tier, under fixed fines per case (a) and under fines proportional to market value and inflation (b), at the doses tested. Labels give the dose as a multiple of the reference. Main setting.

by 27–71%. The honest summary is that the qualitative results follow from the firms' decision rule, which Section 4 describes. The artificial market supplies the financing gains that set the magnitudes. Its feedback strengthens fixed fines, narrows the gap between the instruments, and has no identifiable net effect on transparency in the main setting.

6.7 Learning firms

We also replaced the NPV rule with tabular Q-learning on realised 60-day accounts: the change in financing fees minus resource costs and fines, scaled by initial value. After fixing what could be fixed — seven options (three sizes each of improvement and inflation, plus maintaining), polynomial step sizes (Even-Dar and Mansour, 2003), a state that carries the firm's own signal tercile, a per-period budget, and contract-scale training worlds — the learners still failed the pre-set coverage test in the post-policy large-firm group. More importantly, two tables trained independently in the same environment inflated in about 69% and 15% of decisions, so the learners' policy ordering cannot be separated from training randomness, and they were excluded from the confirmatory experiment.

A diagnosis with randomly assigned actions points to the size of the signal. At the calibrated premium, the median value of the best action over maintaining is about 0.07 basis points of initial market value (0.52 at the 90th percentile). Within a state, the discounted reward over the following 20 periods, the horizon over which an action's financing effects arrive, has an SD of about 190 basis points. As a rough diagnostic, a two-sample test with that noise would need about 1.2×10^8 samples per state and action to tell the median value difference apart from zero, and about 2×10^6 at the 90th percentile, against 10^3–10^4 in our training archives. These figures indicate orders of magnitude; they are not the sample requirement of any particular learning algorithm. Even the one-period reward of maintaining, net of common shocks, has an SD of 2.1 basis points. In addition, the state does not record improvements in progress, so one-step updates credit a delayed financing gain to whatever action happens to be taken when it arrives.

A pre-registered redo. One hypothesis is that a learner of this kind picks up the quick payoff of inflation more easily than the delayed payoff of improvement, and so drifts towards greenwashing. We tested it in a further pre-registered experiment. The design was written into the research contract (Section 23) and committed (`8bb5d9e`) before implementation and runs; the frozen configuration (experiment ID `b4430136...6062`) was committed (`42ec591`) before the confirmatory run. Every scenario was trained ten times independently on new seeds, in two versions of the state: A, the state above, and V, which replaces the least informative feature, the cost tier, with the firm's cumulative real improvement, including improvement still in progress, so that improvement shows in the state as inflation does. In each replicate a pre-policy table is trained under the baseline, and post-policy tables for B, I, F and IF are trained from it; frozen tables are evaluated on five validation seeds and compared with an NPV firm restricted to the same action grid (discrete NPV). Five primary questions were tested across the ten replicates with the Holm correction. All 540 jobs completed. Two infrastructure interruptions were resumed on the same seeds, and the interrupted attempts are kept in the archive.

Results of the redo. Coverage still fails: 91 of 100 tables pass the revised test, and all nine failures are post-policy tables, eight of them in the large-firm group, so the acceptance condition for learning firms is still not met. Training randomness dominates: in version A, the SD of the greenwashing share between tables is 14–18 times its SD within a table across validation seeds. One primary question is supported. Under the information policy, learning firms inflate in 23.7% of decisions against 3.8% for the discrete NPV firm, a difference of +19.9 percentage points [+7.2, +32.6] (Holm $p = 0.031$). About two thirds of this inflation occurs where the discrete NPV firm would not inflate and would almost always improve instead; such decisions are 15.5% of all decisions under I. Their median loss against the NPV choice is 0.042 basis points of market value, far below the one-period reward noise of about 2 basis points. This is a trap the learner cannot learn its way out of, not evidence that it has learned that greenwashing pays. The other four questions are unresolved. Making improvement visible lowers the share of such decisions under I from 15.5% to 9.3% (Holm $p = 0.47$) and does not significantly change the true change under the baseline (+0.003 SD). Learning firms' response to information, +0.157 SD [+0.040, +0.274], is not significant after the correction (Holm $p = 0.057$), and their response to the quadrupled fine is +0.001 SD. In an auxiliary, uncorrected comparison, version V greenwashes 34.7 percentage points less than version A on pre-policy tables trained from scratch, but the decisions it saves become maintaining rather than real improvement. When the five replicates that contain a failing table are dropped, all five questions are unresolved, P1 included.

This is a statement about this learner in this calibration, not about real firms. Firms can value ESG through forward-looking appraisal, as our NPV firms do, and through information other than their own accounts. On ten independently trained tables, the redo confirms what one table suggested: learners trained on realised accounts keep inflating where the NPV rule would not, because the loss from doing so is too small to detect. It does not support the mechanism that invisible improvement causes the trap. Because coverage still fails and behaviour depends mainly on which table was trained, learning firms remain outside the main experiment.

7 Robustness and limitations

Appendix A collects the three main contrasts across all 17 settings we ran (Table 10). The ordering I > F > B holds in all of them, and excluding runs that triggered the price guard changes no verdict. The size of the information advantage ranges from +0.003 to +0.224 SD and depends mainly on two assumptions: that real improvement lasts, and that a fine is charged once per case. A pre-registered test on

new seeds that changes both together keeps the ordering in all four settings, with an advantage of +0.004 to +0.012 SD. The main result is therefore an ordering that is robust across the settings tested, with a magnitude that is not. The remaining limitations are these.

Costs and scope of policy. Neither the cost of transparency nor that of audits is modelled, so the comparison is not about social cost. In the first experiment the regulator does not choose whom to audit and audit intensity is fixed at 1% per period; Section 8 tests two targeted rules and a doubled budget. Section 9 adds assurance providers; there are no rating agencies.

The financing channel. With the calibrated taste, the local response of a firm's financing rate to its signal — the quantity firms decide on — is about 2.2 basis points per SD, an order of magnitude below the empirical cost-of-equity association, because of the reduced-form financing contract. We did not adjust the contract to close the gap, since its parameters are not estimated.

Firm rationality. Firms are myopic NPV maximisers. Learning firms failed the coverage test in the first comparison and in 9 of 100 tables of the pre-registered redo, and their behaviour depends mainly on which table was trained (Section 6.7), so the main results say nothing about boundedly rational firms.

Data. The CSI 300 contains only large firms and is observed at its current composition. A rating is a proxy, not the true attribute, and the date on a rating need not be the date it became public.

Market layer. In a qualification audit of 60 fixed worlds, 57 passed and 3 failed mechanical checks; 37 confirmatory runs triggered the price guard; and the simulated-moments fit leaves lag-1 return autocorrelation too negative (−0.18 against −0.03). All are kept in the record, and the main contrasts do not change when the affected runs are excluded.

8 Targeted audits: a second pre-registered experiment

In every run so far the regulator audits each firm with probability 1% per period. Two results of the first experiment make the allocation of audits worth testing. The statutory fine is negligible for large firms, which inflate in 66% of their decisions, and a firm with an open case continues to inflate whatever the fine (Section 4). Unlike the fine, the audit probability sets both how long inflation survives (A_w) and the discounted value of paying on detection (D), so it also reaches firms that continue. We therefore ask whether auditing some firms more and others less, at an unchanged expected number of audits, does better than random audits, and whether it matters that firms know the rule.

Design. The design was written into the research contract (Section 21) and committed (`962d1ee`) before implementation and runs; the pipeline checks, the frozen configuration (experiment ID `06bec148...2a32`) and the analysis script were committed (`3fced38`) before the confirmatory run. The experiment uses 30 new seeds, 42001–42030, and reruns B, I and F on them. On each audit date firm i is audited if a uniform draw falls below its audit probability q_i. The draws are shared across scenarios, and every targeted rule keeps $\sum_i q_i = 2$, the expected number of audits when each of the 200 firms is audited with probability 1%. The rules are:

- *Random* (B): $q_i = 1\%$.

Table 6: Primary questions of the targeted-audit experiment

Question	Contrast	Mean (SD)	95% interval	Holm p
E1 Value targeting	V − B	−0.191	[−0.222, −0.159]	8×10^{-13}
E2 Jump targeting	J − B	−0.136	[−0.156, −0.116]	8×10^{-14}
E3 Evasion	J − J_{hidden}	−0.131	[−0.148, −0.114]	4×10^{-15}
E4 Size divide	V − B, large − small	+0.617	[+0.551, +0.683]	3×10^{-17}
E5 Information vs. targeted fine	I − VF	+0.156	[+0.127, +0.186]	1×10^{-11}

Notes: Main setting (natural decay 0.2 SD per year), policy window, 30 paired seeds (42001–42030). E3 was predicted to be negative and E4 positive; E1, E2 and E5 had no predicted sign. No run in these scenarios triggered the price guard.

- *By value* (V): q_i is proportional to market value. At the initial distribution of values the median small firm is audited with probability 0.41% per period, the median large firm 3.0% and the largest firm 8.0%.
- *By jump* (J): half the budget is spread evenly ($q_i = 0.5\%$) and the other half is shared among flagged firms, those whose report (true level plus inflation, before the transparency discount) rose by more than $h = 0.125$ SD, half the per-period cap, since the previous audit date. The random half leaves no firm unaudited.
- *Doubled random audits* (A2): $q_i = 2\%$, twice the budget, as a reference for auditing more.

When a rule is public, firms decide with their own audit probability: under V with the day's q_i, under J by predicting whether new inflation will be flagged and choosing the better of staying below the threshold and accepting the flag. When it is hidden (V_{hidden}, J_{hidden}), firms decide as in the first experiment, so targeting changes only who is caught. Firms under J ignore the effect of a flag on older undetected inflation, which understates the cost of being flagged and therefore evasion. The 15 scenarios are B, I, F, A2, V, V_{hidden}, J, J_{hidden} and VF (value targeting with the fixed fine at four times the statutory maximum) in the main setting, and B, I, V, J, J_{hidden} and VF without natural decay. Five primary questions were tested two-sided with the Holm correction (Table 6), and five analytical predictions derived from the propositions were written down with the design; all five held. All 450 runs completed; five triggered the price guard, all on one seed at day 3101 and none in a scenario of the primary questions.

Results. At an equal number of audits, both targeted rules do worse than random audits when firms know the rule (E1, E2), although both find more: 50% of audits under V and 69% under J find inflation, against 28% under random audits (Table 7). Hidden, neither rule differs significantly from random audits in the equal-weighted true level: V_{hidden} − B is −0.005 SD [−0.012, +0.001] and J_{hidden} − B is −0.005 [−0.011, +0.001]; intervals that include zero do not establish equivalence. The loss therefore appears to come mainly from firms' response to a known rule rather than from detection itself; publishing the jump rule costs 0.131 SD (E3).

Value targeting moves greenwashing from large to small firms. Large firms' share of inflating decisions falls from 64% to 47% and small firms' rises from 5% to 35%. The size divide reverses: the large-minus-small difference in the change of the true level moves from −0.30 to +0.32 SD (E4). Weighted by initial market value, the true level rises (+0.053 SD [+0.043, +0.063]), but small firms are 60% of all firms, and the equal-weighted level falls. Small firms lose more deterrence than large firms gain. The fine is large relative to small firms' value and small relative to large firms', and the detection discount is strongly concave in the audit probability ($D = 0.455$ at 1% and 0.87 at 8%). Measured by the expected discounted fixed fine relative to value, with the average firm under random audits at 1, value targeting lowers the median small firm from 2.46 to 1.37 but raises the median large firm only from 0.34

Table 7: Targeted-audit experiment: outcomes and audits by scenario

Scenario	True change	Inflating decisions (%)		Hit rate	Audits received (%)
	(SD)	all	large / mid / small	(%)	large / mid / small
B Random	−1.193	20.3	64 / 37 / 5	28	9 / 32 / 59
I Information, $\tau = 0.6$	−0.875	0	0 / 0 / 0	18	9 / 32 / 59
F Fixed fine 4×	−1.066	11.6	53 / 17 / 2	19	9 / 32 / 59
A2 Doubled random	−1.075	13.3	54 / 23 / 2	21	10 / 31 / 59
V By value	−1.384	37.2	47 / 38 / 35	50	35 / 41 / 23
V_{hidden}	−1.198	22.2	69 / 39 / 6	54	36 / 42 / 23
J By jump	−1.329	32.5	75 / 56 / 14	69	13 / 42 / 45
J_{hidden}	−1.198	21.2	65 / 39 / 5	65	16 / 45 / 39
VF Value, fixed fine 4×	−1.031	10.8	28 / 16 / 5	32	36 / 42 / 23

Notes: Means over 30 seeds; main setting, policy window. The hit rate is the share of audits that find inflation, and audits received is each size tier's share of all audits; the three tiers are 10%, 30% and 60% of firms. B, I and F are rerun on the new seeds, so their values differ slightly from Table 4. True change is negative in every scenario because of the assumed decay; only differences between rows are interpretable.

to 0.53. Nor does it end large firms' open cases. By Proposition 2, with A_w evaluated at the firm's own audit probability, a firm with an open case stops inflating at $\tau = 0.1$ only if that probability is at least 10.5% per period. All 81,253 continuing inflations under V came from firms below that rate, and the 313 firm-periods that reached it had none.

Firms evade the jump rule. When the rule is public, 44% of new inflations are flagged at the next audit date, against 54% when it is hidden (a difference of −9.6 percentage points [−10.9, −8.2]; −13.8 points without decay): firms keep inflation below the threshold. Unflagged firms then face an audit probability of only 0.5%, and small firms, which the fine deterred under random audits, open more new cases; their share of inflating decisions rises by 8.7 points against J_{hidden}.

Auditing more works where auditing differently does not. Doubling random audits raises the true level by +0.118 SD [+0.103, +0.133] and lowers inflation in every size tier, but large firms still inflate in 54% of their decisions against 2% for small firms. Both targeted rules fall far short of it (V − A2 is −0.309 SD and J − A2 is −0.254). Information still beats targeted enforcement with a heavier fine: I − VF is +0.156 SD (E5). Adding value targeting to the quadrupled fine helps little in the equal-weighted level (VF − F is +0.034 [+0.025, +0.044]) and more when weighted by value (+0.181 [+0.160, +0.203]). On the new seeds the main comparison of the first experiment replicates: I − F is +0.191 SD [+0.165, +0.216], against +0.205 in Table 3. Without natural decay all effects are small and keep their signs (V − B is −0.003 SD, J − B −0.008 and I − VF +0.026), so magnitudes are again read as the range between no decay and 0.2 SD per year (Section 6.1).

Scope. The result concerns the two rules tested: a proportional rule, and a jump rule with one threshold, a one-period window and half the budget random. No rule was optimised, so the experiment does not show that targeting cannot beat random audits, only that these two rules do worse once firms know them. The mechanism is reallocation under a fixed budget. Auditing large firms or recent jumps more means auditing small firms or quiet old inflation less, and firms that know the rule inflate where audits thin out. The experiment keeps the other assumptions of the main setting: per-case booking, a common washing cost ratio, uniform decay and fixed doses.

9 Assurance providers: a third pre-registered experiment

The information instrument of the first experiment is an abstract transparency τ. One institution that makes inflated reports easier to see through is third-party assurance, and it has a conflict that transparency does not: the firm pays the provider and, unless providers are assigned, chooses it. The 2024 exchange guidelines do not require assurance, although a firm that hires a provider must disclose the provider's independence and qualifications and its assurance report (Shanghai Stock Exchange, 2024, Art. 58). Under Article 213(3) of the Securities Law, a securities service provider that fails to exercise due diligence and whose documents contain false records, misleading statements or material omissions forfeits its business income and is fined one to ten times that income, or CNY 0.5–5 million if it has no business income or less than CNY 0.5 million (Standing Committee of the National People's Congress, 2019, Art. 213(3)). We use the upper limit of that low-income band, CNY 5 million, as the benchmark liability and assume that the provision applies to ESG assurance; we make no judgement on whether it does. We therefore add strategic assurance providers and ask how much assurance does for the true level, and how this depends on the provider's liability, on who chooses the provider and on which firms must be assured.

Design. The design was written into the research contract (Section 22) and committed (`307168c`) before implementation and runs; the pipeline checks, the frozen configuration (experiment ID `35c4f767...339a`) and the analysis script were committed (`61b1205`) before the confirmatory run. The experiment uses 30 new seeds, 43001–43030, and NPV firms; policy switches at day 1201 as before. Four providers inspect each assured firm once per period, before its report is published. A provider of quality $a_k \in [0, 1]$ finds the inflation in a client's report with probability a_k and corrects all of it before publication; there is no fine and nothing is made public. The firm pays a fee for each inspection. At the start of each assurance year (240 days) each provider sets its quality as a myopic best response, $a_k = \min(1, \lambda s_k)$ with $\lambda = L\bar{q}/(2\kappa)$. This minimises the effort cost κa^2 plus the expected liability, where s_k is the share of its inspections in the previous year that met an inflated report, $\bar{q} = 1\%$ is the regulator's audit probability, and L is the liability the provider pays when the regulator later detects inflation in a report it inspected. Liability doses are 0, 1, 4, 10, 16 and 100 times the CNY 5 million benchmark. The fee and the effort cost κ are assumptions, so doses are read relative to κ. Providers are either assigned to firms at random (R) or chosen by each firm at the start of every year (F), and assurance is mandatory for all firms (ALL), only for the 20 large firms (LRG), or voluntary (VOL16). The 18 scenarios are B, I, VOL16, ALL_R0 to ALL_R100, ALL_F1, ALL_F16, LRG_R16 and LRG_F16 in the main setting, and five of them without natural decay. From Proposition 2, a quality of 0.0827 is enough to stop every firm with an interior solution from inflating at the baseline transparency: inspection comes every period and before publication, whereas the regulator audits 1% of firms per period. Five primary questions were tested two-sided with the Holm correction (Table 8), and six analytical predictions were written down with the design; all six held. All 540 runs completed. Twenty-nine triggered the price guard; as pre-registered, they remain in the main analysis, and a sensitivity analysis that excludes them leaves all five primary verdicts unchanged.

Results. With strong liability and assigned providers, assurance works. At 16 times the benchmark liability, ALL_R16 raises the true level by +0.273 SD over the baseline (C1), and the share of inflating decisions falls from 19.0% to 6.2%, in all three size tiers. Letting firms choose their provider removes most of that gain: ALL_F16 is 0.199 SD below ALL_R16 (C2), although in scenario means it remains about 0.07 SD above the baseline (Table 9), and 21.0% of decisions inflate, slightly more than without assurance. Counted by decisions, 96.9% of assured firms' inflating decisions (46,489 of 47,953) were made by clients of the year's lowest-quality provider. The mechanism is an escape cycle. The provider

Table 8: Primary questions of the assurance experiment

Question	Contrast	Mean (SD)	95% interval	Holm p
C1 Strong liability	ALL_R16 − B	+0.273	[+0.234, +0.312]	5×10^{-14}
C2 Firm choice	ALL_F16 − ALL_R16	−0.199	[−0.233, −0.165]	4×10^{-12}
C3 Benchmark liability	ALL_R1 − B	+0.046	[+0.035, +0.058]	8×10^{-9}
C4 Size divide	LRG_R16 − B, large − small	+0.563	[+0.516, +0.610]	3×10^{-20}
C5 Information vs. ALL_R16	I − ALL_R16	+0.028	[+0.018, +0.038]	3×10^{-6}

Notes: Main setting (natural decay 0.2 SD per year), policy window, 30 paired seeds (43001–43030). Intervals are paired 95% t intervals, not adjusted for multiple comparisons; the Holm correction applies to the p-values. C4 is the difference between large and small firms in the change of the true level. C1, C2 and C4 had a predicted sign; C3 and C5 did not. The number in a scenario name is the liability as a multiple of the CNY 5 million benchmark; R means assigned providers and F providers chosen by firms.

whose clients inflated has a high s_k and so a high quality the next year, and at the start of that year those firms move to a provider whose clients were clean and whose quality is therefore low; about 46 firms switch each year.

At the benchmark liability (ALL_R1), provider quality averages only 0.014 and the gain is small, +0.046 SD (C3). This is the range in which the analytical benchmark predicted that weak assurance acts as insurance against the fine: corrected inflation can no longer be fined, and among firms that have no undetected inflation when they decide, the share of decisions that inflate rises by 0.9 percentage points [+0.7, +1.2]. The share of all inflating decisions rises from 19.0% to 20.0%. The true-level change nevertheless improves relative to the baseline, because improving firms improve by more (0.0229 against 0.0209 SD per improvement on average); this decomposition was made after the results and its mechanism was not tested separately. Against the baseline the dose response is +0.000, +0.046, +0.145, +0.256, +0.273 and +0.232 SD at 0, 1, 4, 10, 16 and 100 times the benchmark. Among the doses tested, 16 times gives the largest gain and 100 times a smaller one; no dose between them was run. At 100 times, quality averages 0.43, yet 8.6% of decisions inflate, more than at 16 times. The pre-registered prediction of a quality cycle was tested under ALL_R16. High quality corrects the stock of inflation, so s_k and with it the next year's quality fall and inflation can return. In ALL_R16 the yearly mean quality first exceeds 0.0827 and later falls below it in 28 of 30 seeds, and some decisions still inflate in the last four assurance years in all 30 seeds. The same feedback may explain the smaller gain at 100 times, but this was not tested separately.

Mandating assurance only for large firms narrows the size divide (LRG_R16, large minus small +0.563 SD; C4). Large firms' share of inflating decisions falls from 64.7% to 24.1%, as when every firm is assured, but mid-sized firms' rises from 34.4% to 38.8% and small firms' from 3.6% to 6.1%, so the equal-weighted gain is only +0.057 SD. Information still beats the assurance setting with the largest gain among those tested (ALL_R16), by a small margin: I − ALL_R16 is +0.028 SD (C5), +0.075 weighted by value, and +0.227 against assurance with firm choice. No firm buys voluntary assurance, so VOL16 is bit-identical to the baseline; investors see only reported scores and do not reward assurance, so this follows from the model's assumptions and does not predict practice. Without natural decay all effects are small and keep their signs (+0.030 for C1, −0.010 for C2, +0.004 for C3 and +0.009 for C5).

Scope. The strength of assurance comes from its timing: an inspection every period before publication, against regulatory audits of 1% per period. What limits it is the provider's incentive, set by liability, and who chooses the provider. Providers respond myopically once a year, do not anticipate losing clients, and know the true incidence of inflation in their clients' reports of the previous year, which overstates what they know. The fee and the effort cost are assumptions, and the benchmark liability assumes that

Table 9: Assurance experiment: outcomes and provider quality by scenario

Scenario	True change	Inflating decisions (%)		Mean	Quality faced
	(SD)	all	large / mid / small	quality	by clients
B No assurance	−1.218	19.0	65 / 34 / 4	—	—
I Information, $\tau = 0.6$	−0.917	0.0	0 / 0 / 0	—	—
ALL_R0	−1.218	19.0	65 / 34 / 4	0	0
ALL_R1	−1.172	20.0	60 / 35 / 6	0.014	0.014
ALL_R4	−1.073	16.6	60 / 30 / 3	0.046	0.046
ALL_R10	−0.962	8.0	35 / 13 / 1	0.080	0.080
ALL_R16	−0.945	6.2	24 / 10 / 1	0.110	0.110
ALL_R100	−0.986	8.6	25 / 15 / 3	0.425	0.425
ALL_F1	−1.184	20.6	64 / 37 / 5	0.010	0.010
ALL_F16	−1.144	21.0	63 / 37 / 6	0.131	0.110
LRG_R16	−1.161	17.7	24 / 39 / 6	0.309	0.309
LRG_F16	−1.182	19.1	42 / 38 / 6	0.297	0.238

Notes: Means over 30 seeds; main setting, policy window. Quality faced by clients weights each provider by its clients; with firm choice it is below the providers' mean because firms leave high-quality providers. VOL16 is identical to B. B and I are rerun on the new seeds. True change is negative in every scenario because of the assumed decay; only differences between rows are interpretable.

Article 213(3) applies to ESG assurance. The model has no provider reputation or exit, no mandatory rotation, no reporting by providers to the regulator and no minimum assurance standard. Nor does it publish whether a firm is assured or the provider's report, although the exchange guidelines require a firm that hires a provider to disclose them; this simplification limits what the voluntary-assurance result says about practice. The result under firm choice suggests that rotation or assignment may matter, but rotation was not tested. The experiment keeps the other assumptions of the main setting: per-case booking, a common washing cost ratio, uniform decay, random audits and a fixed transparency dose.

10 Conclusion

In an artificial market calibrated to Chinese magnitudes, raising the transparency of ESG reports did more for firms' true ESG level than raising fixed fines. This held at the doses tested and in every setting we ran: both pre-registered decay rates, three intermediate rates, mean-reverting true levels, a fine for every false report and firm-specific washing costs, and in a pre-registered test that combined reversion with a fine for every report. It held again on the new seeds of the second experiment, where transparency also beat value-targeted audits combined with the quadrupled fine, and in the third, where it beat the assurance setting with the largest gain among those tested (ALL_R16) by a small margin. How much more depends on two things the model cannot settle. If real improvements fade towards the mean, as the only rating panel we hold suggests, transparency of 0.6 no longer stops greenwashing and its advantage over a quadrupled fine nearly vanishes. If every false report is fined separately, fines lose their deterrence plateau and the gap narrows to a quarter of its main-setting size. With both, transparency still comes first, but by +0.004 to +0.012 SD. Whether fines are booked per case or per report is a question of law and enforcement practice. Whether ESG performance reverts is a question for verified rating data. Both deserve answers before either instrument is preferred on these grounds.

Some results hold across these settings. None of the instruments works without investors who value ESG, and the market's own drive to recruit them is weak. A fine of the size the Securities Law allows is negligible for large firms, where greenwashing concentrates. A fine proportional to value reverses that tilt even at the lowest dose tested, but at the equal-amount reference it leaves 62–84% of decisions inflating.

And with heterogeneous firms, transparency works firm by firm rather than through a single threshold.

The second experiment adds a caution about targeting audits. At an unchanged number of audits, auditing large firms more or auditing firms whose reports jump did worse than random audits once firms knew the rule, because greenwashing moved to where audits thinned out; with the rule hidden, targeting made no statistically significant difference, and doubling random audits helped. A rule that catches more inflation per audit can still deter less, so a targeting rule has to be judged by how firms respond to it.

The third experiment points the same way for assurance, one of the institutions behind transparency. Mandatory assurance helped when providers faced strong liability and were assigned. Letting firms choose their own provider undid most of that gain, as their greenwashing concentrated among clients of the laxest provider, and at the benchmark liability providers had little reason to look. In the model, how much assurance achieves depends on the provider's liability and on who chooses the provider.

The model is a laboratory, not a forecast. Its value lies in making each step from assumption to result inspectable: the analytical benchmark shows which results follow from the firm's decision rule, the feedback decomposition shows how much the market adds, and the pre-registration and archive show what was decided before the results were seen. Verified rating panels, audit rules designed against firms' responses, the rotation and reputation of assurance providers, and firms that learn with enough data are the natural next steps.

References

W. Brian Arthur, John H. Holland, Blake LeBaron, Richard Palmer, and Paul Tayler. Asset pricing under endogenous expectations in an artificial stock market. In W. Brian Arthur, Steven N. Durlauf, and David A. Lane, editors, *The Economy as an Evolving Complex System II*, pages 15–44. Addison-Wesley, 1997.

Rudolf Avenhaus, Bernhard von Stengel, and Shmuel Zamir. Inspection games. In Robert J. Aumann and Sergiu Hart, editors, *Handbook of Game Theory with Economic Applications*, volume 3, chapter 51, pages 1947–1987. Elsevier, 2002. doi: 10.1016/S1574-0005(02)03014-X.

Gary S. Becker. Crime and punishment: An economic approach. *Journal of Political Economy*, 76(2): 169–217, 1968. doi: 10.1086/259394.

Florian Berg, Julian F. Kölbel, and Roberto Rigobon. Aggregate confusion: The divergence of ESG ratings. *Review of Finance*, 26(6):1315–1344, 2022. doi: 10.1093/rof/rfac033.

Yi-Chun Chen, Mingyi Hung, and Yongxiang Wang. The effect of mandatory CSR disclosure on firm profitability and social externalities: Evidence from China. *Journal of Accounting and Economics*, 65 (1):169–190, 2018. doi: 10.1016/j.jacceco.2017.11.009.

Hans B. Christensen, Luzi Hail, and Christian Leuz. Mandatory CSR and sustainability reporting: Economic analysis and literature review. *Review of Accounting Studies*, 26(3):1176–1248, 2021. doi: 10.1007/s11142-021-09609-5.

Magali A. Delmas and Vanessa Cuerel Burbano. The drivers of greenwashing. *California Management Review*, 54(1):64–87, 2011. doi: 10.1525/cmr.2011.54.1.64.

Alexander Dyck, Adair Morse, and Luigi Zingales. How pervasive is corporate fraud? *Review of Accounting Studies*, 29(1):736–769, 2024. doi: 10.1007/s11142-022-09738-5.

Eyal Even-Dar and Yishay Mansour. Learning rates for Q-learning. *Journal of Machine Learning Research*, 5:1–25, 2003.

J. Doyne Farmer and Duncan Foley. The economy needs agent-based modelling. *Nature*, 460(7256): 685–686, 2009. doi: 10.1038/460685a.

Peter Fiechter, Jörg-Markus Hitz, and Nico Lehmann. Real effects of a widespread CSR reporting mandate: Evidence from the European Union's CSR Directive. *Journal of Accounting Research*, 60(4): 1499–1549, 2022. doi: 10.1111/1475-679X.12424.

Allen Goss and Gordon S. Roberts. The impact of corporate social responsibility on the cost of bank loans. *Journal of Banking & Finance*, 35(7):1794–1810, 2011. doi: 10.1016/j.jbankfin.2010.12.002.

Volker Grimm, Steven F. Railsback, Christian E. Vincenot, Uta Berger, Cara Gallagher, Donald L. DeAngelis, Bruce Edmonds, Jiaqi Ge, Jarl Giske, Jürgen Groeneveld, Alice S. A. Johnston, Alexander Milles, Jacob Nabe-Nielsen, J. Gareth Polhill, Viktoriia Radchuk, Marie-Sophie Rohwäder, Richard A. Stillman, Jan C. Thiele, and Daniel Ayllón. The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, 23(2):7, 2020. doi: 10.18564/jasss.4259.

Robert Heinkel, Alan Kraus, and Josef Zechner. The effect of green investment on corporate behavior. *Journal of Financial and Quantitative Analysis*, 36(4):431–449, 2001. doi: 10.2307/2676219.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6 (2):65–70, 1979.

Eun-Hee Kim and Thomas P. Lyon. Greenwash vs. brownwash: Exaggeration and undue modesty in corporate sustainability disclosure. *Organization Science*, 26(3):705–723, 2015. doi: 10.1287/orsc.2014.0949.

Ralph S. J. Koijen and Motohiro Yogo. A demand system approach to asset pricing. *Journal of Political Economy*, 127(4):1475–1515, 2019. doi: 10.1086/701683.

Blake LeBaron. Agent-based computational finance. In Leigh Tesfatsion and Kenneth L. Judd, editors, *Handbook of Computational Economics*, volume 2, pages 1187–1233. Elsevier, 2006. doi: 10.1016/S1574-0021(05)02024-1.

Thomas P. Lyon and John W. Maxwell. Greenwash: Corporate environmental disclosure under threat of audit. *Journal of Economics & Management Strategy*, 20(1):3–41, 2011. doi: 10.1111/j.1530-9134.2010.00282.x.

Thomas P. Lyon and A. Wren Montgomery. The means and end of greenwash. *Organization & Environment*, 28(2):223–249, 2015. doi: 10.1177/1086026615575332.

Christopher Marquis, Michael W. Toffel, and Yanhua Zhou. Scrutiny, norms, and selective disclosure: A global study of greenwashing. *Organization Science*, 27(2):483–504, 2016. doi: 10.1287/orsc.2015.1039.

Benjamin A. Olken. Promises and perils of pre-analysis plans. *Journal of Economic Perspectives*, 29(3): 61–80, 2015. doi: 10.1257/jep.29.3.61.

Ľuboš Pástor, Robert F. Stambaugh, and Lucian A. Taylor. Sustainable investing in equilibrium. *Journal of Financial Economics*, 142(2):550–571, 2021. doi: 10.1016/j.jfineco.2020.12.011.

Lasse Heje Pedersen, Shaun Fitzgibbons, and Lukasz Pomorski. Responsible investing: The ESG-efficient frontier. *Journal of Financial Economics*, 142(2):572–597, 2021. doi: 10.1016/j.jfineco.2020.11.001.

Karl H. Schlag. Why imitate, and if so, how? A boundedly rational approach to multi-armed bandits. *Journal of Economic Theory*, 78(1):130–156, 1998. doi: 10.1006/jeth.1997.2347.

Shanghai Stock Exchange. Guidelines No. 14 of Shanghai Stock Exchange for Self-Regulation of Listed Companies—Sustainability Report (Trial), 2024. URL https://english.sse.com.cn/news/newsrelease/c/c_20240412_10753174.shtml. Issued 12 April 2024, effective 1 May 2024. Article 3 sets the companies required to report, Article 61 provides self-regulatory measures or disciplinary sanctions for violations, and Article 63 requires the first reports, for 2025, by 30 April 2026.

Standing Committee of the National People's Congress. Securities Law of the People's Republic of China (2019 Revision), 2019. URL https://www.samr.gov.cn/zw/zfxxgk/fdzdgknr/bgt/art/2023/art_725883cb6dda4fdf965a3d14358e279f.html. Revised 28 December 2019, effective 1 March 2020.

Peter D. Taylor and Leo B. Jonker. Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2):145–156, 1978. doi: 10.1016/0025-5564(78)90077-9.

Lingxiao Zhao, Maria Polukarov, and Carmine Ventre. Agent-based analysis of green disclosure policies and their market-wide impact on firm behavior. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*, pages 2355–2363, Detroit, MI, 2025. IFAAMAS. doi: 10.65109/nnvi4131.

Lingxiao Zhao, Maria Polukarov, and Carmine Ventre. Green disclosure policies and market dynamics: Evidence from agent-based ESG models. *Autonomous Agents and Multi-Agent Systems*, 40(1):6, 2026. doi: 10.1007/s10458-026-09734-y.

Yang Zhao, Wenrui Zeng, and Zhongbo Jing. Can ESG performance enhance corporate valuation? Empirical evidence from the cost of equity capital. *Financial Market Research*, pages 1–14, 2024. URL https://www.nafmii.org.cn/yj/jrscyj/qk/2024/202405/. Issue 5 (total issue 144). In Chinese.

A Robustness

Table 10 reports the three main contrasts in every setting we ran; Section 7 summarises them. The subsections below describe the settings that change the size of the information advantage or the shape of the response to transparency.

A.1 Mean reversion (exploratory)

The evidence on decay does not support uniform decay. In a panel of Huazheng ratings of all A-share firms for 2009–2022, which we obtained from a secondary source and have not yet verified against an official export, the average rating has no trend (yearly means between 3.94 and 4.32 on a 1–9 scale), and ratings revert towards the middle: the mean next-quarter change is +0.30 notches for the lowest grade

Table 10: Policy contrasts across all settings (true change over the policy window, SD)

Setting	Status	I − B	F − B	I − F
Uniform decay 0.2 SD/year (main)	Pre-registered	+0.331	+0.125	+0.205
No decay	Pre-registered	+0.040	+0.005	+0.035
Uniform decay 0.05 SD/year	Exploratory	+0.073	+0.012	+0.061
Uniform decay 0.1 SD/year	Exploratory	+0.130	+0.030	+0.100
Uniform decay 0.15 SD/year	Exploratory	+0.217	+0.071	+0.146
Mean reversion, $\kappa = 0.32$	Exploratory	+0.027	+0.021	+0.006
Mean reversion, $\kappa = 0.50$	Exploratory	+0.021	+0.018	+0.003
A fine for every false report	Exploratory	+0.216	+0.158	+0.058
Firm-specific washing cost	Exploratory	+0.320	+0.118	+0.202
Reversion $\kappa = 0.32$ + fine per report	Joint test	+0.055	+0.043	+0.012
Reversion $\kappa = 0.50$ + fine per report	Joint test	+0.043	+0.037	+0.006
with firm-specific washing cost, $\kappa = 0.32$	Joint test	+0.054	+0.043	+0.010
with firm-specific washing cost, $\kappa = 0.50$	Joint test	+0.041	+0.037	+0.004
Equal cost multipliers	Pre-registered	+0.362	+0.146	+0.216
Size–ESG independence	Pre-registered	+0.350	+0.126	+0.224
Book-equity dispersion	Pre-registered	+0.269	+0.089	+0.180
Main setting, new seeds	Second experiment	+0.318	+0.127	+0.191

Notes: Means of 30 paired seeds; every paired 95% interval excludes zero. Rows other than the decay rows use 0.2 SD per year of uniform decay, except the mean-reversion rows, which replace it. Exploratory rows were designed after the confirmatory results and pre-registered before their own runs (contract Sections 20.16 and 20.21). Joint-test rows are confirmatory, pre-registered in contract Section 24 before implementation and runs, on new seeds 44001–44030; their I − F and F − B are the eight primary contrasts, all with Holm $p < 2 \times 10^{-6}$. Equal cost multipliers, size–ESG independence and book-equity dispersion are the initialisation controls declared before any policy result. The last row reruns B, I and F on the 30 seeds of the targeted-audit experiment (Section 8).

and −0.48 for the highest. Mean reversion is a different functional form from uniform decay and is not covered by the range above. The panel may not settle the level either: if ratings are relative to peers, an economy-wide trend cannot be seen in them.

We therefore added a mean-reverting form: the true level reverts towards the day's cross-sectional mean at rate κ per year, with idiosyncratic shocks that keep the cross-sectional variance at one, and NPV firms account for the reversion when they value improvement. The panel's quarterly autocorrelation of 0.88 implies $\kappa = 0.50$; removing white-noise measurement error with the ratio of the four-quarter to the one-quarter autocorrelation gives $\kappa = 0.32$. We treat both as data-motivated sensitivity values, not as estimates or bounds of how fast the true level reverts: the panel is unverified, and its reversion mixes measurement noise, rating discreteness and firms' own actions, whose net effect on these numbers we cannot sign. Under both rates the ordering I > F > B survives, but the advantage of information almost disappears. The gap between the instruments is +0.006 SD [+0.004, +0.008] at $\kappa = 0.32$ and +0.003 [+0.002, +0.004] at $\kappa = 0.50$, and the effect of information against the baseline is +0.027 and +0.021. Both lie below the lower end of the uniform-decay range by amounts the paired intervals resolve. The fixed-fine effect, +0.021 and +0.018, stays inside it. The mechanism is the one Proposition 2 predicts. Reversion erodes real improvement (half-lives of 1.4–2.2 years) while inflation is removed only by audits, so A_r falls from 3.73 to 2.01 and 1.51 years, and the threshold at $w = 0.2$ rises to 0.77 and 0.82. Transparency of 0.6 no longer stops inflation: 20–22% of decisions still inflate under I, against 25–28% under B, so transparency acts much like a heavier fine. Reversion is thus not a decay rate between the two ends of the range. Like no decay, it does not pull the average down; like strong decay, it makes room for inflation; and it lowers the value of real improvement.

A.2 A fine for every false report (exploratory)

When every false report is fined separately from day 0, and NPV firms charge each new inflation its expected fine, the share of inflating decisions falls from 14.3% at the statutory fine to 6.6%, 2.4% and 0.8% at 2, 4 and 8 times it. Under per-case booking the corresponding shares are 22.3%, 15.5%, 13.3% and 12.5%. Deterrence still diminishes: each doubling adds +0.094, +0.064 and +0.034 SD, and each step is smaller than the last by an amount the paired intervals resolve. It diminishes because each doubling deters fewer and larger firms, for which the fine is a smaller share of value; the remaining inflation comes from the largest firms. The fine becomes a stronger instrument (+0.158 SD against the baseline, from +0.125), the baseline itself improves, and information is unaffected because it stops inflation outright. The gap between information and the quadrupled fine therefore shrinks from +0.205 to +0.058 SD [+0.042, +0.074], still positive. Which booking is right is a legal question: several years of false disclosure are often handled in a single penalty decision, but a separate fine per false report is also a plausible reading of the law.

A.3 Firm-specific washing costs (exploratory)

We drew each firm's ratio w_i from a lognormal distribution with mean 0.2 and log-SD $\ln 2$ (5th to 95th percentile 0.05 to 0.49). Inflation then no longer stops at a common threshold. At $\tau = 0.6$, 7.3% of decisions still inflate (none with a common ratio), in every seed. At $\tau = 0.8$, 0.4% do, in 29 of 30 seeds. The decline is governed firm by firm: all 53,015 inflating decisions after the switch, across the three transparency levels, came from firms whose own threshold $\tau^*(w_i)$ exceeded the current τ, as Proposition 2 implies. The policy comparison barely moves. Information beats the baseline by +0.320 SD (common ratio +0.331) and the fine by +0.202 (+0.205). The plateau above $\tau = 0.6$ in Figure 3 partly reflects the common ratio: with dispersed ratios, raising transparency from 0.6 to 0.8 still cuts inflation from 7.3% to 0.4%.

A.4 Both assumptions together (pre-registered)

Both assumptions narrow the gap, mean reversion by weakening transparency and per-report fines by strengthening the fixed fine, so together they could have let the quadrupled fine catch up with transparency. It did not, because a fine for every report also applies under I, which keeps the baseline fine: inflation under I falls from 18–21% of decisions with reversion alone to 0.8–1.3%, and under F to 2.2–2.4%. Proposition 2 implies the same ordering of who inflates: with a fine per report, a firm that inflates under I in a given state would also inflate under F, as long as its choice is not at the per-period cap. The advantage of transparency is +0.012 SD [+0.009, +0.015] at $\kappa = 0.32$ and +0.006 [+0.005, +0.008] at $\kappa = 0.50$; firm-specific washing costs lower it by about 0.002. The reference blocks of the same run reproduce the earlier exploratory values on the new seeds (main setting +0.200, fine per report +0.067, reversion alone +0.006 and +0.002).

What remains untested. The mean-reversion rates are data-motivated sensitivity values from an unverified panel, not estimates or bounds; rates between 0 and 0.32, reversion combined with decay, and firms that ignore reversion were not run. The per-report booking was tested only in the main setting and together with mean reversion.

B Reproducibility

All code, configurations and evidence are in the public repository https://github.com/kanzaler123/financial-ABM (MIT licence). Vendor data without a redistribution licence (Huazheng ratings, market data) are not included; collection scripts and hash manifests are.

- *Pre-registration:* research contract, Section 20 (`docs/disclosure_research_contract.md`), committed before the confirmatory run (`2b08f65`); supplements in Sections 20.16–20.21.
- *Confirmatory experiment:* configuration `configs/disclosure_stage_f_confirmatory_v1.json`, experiment ID `90ee8606...5d57`, seeds 41001–41030; report `reports/disclosure_stage_f_20261014/`.
- *Supplements* (under `reports/`): `disclosure_robustness_20261018/` (contract Section 20.21), `disclosure_decay_grid_20261016/`, `disclosure_feedback_counterfactual_20261016/`, `disclosure_feedback_strong_20261016/`, `disclosure_feedback_channels_20261017/`, `disclosure_feedback_channels_main_20261017/`.
- *Targeted audits:* research contract, Section 21, committed before implementation and runs (`962d1ee`); pipeline checks, frozen configuration and analysis script committed before the confirmatory run (`3fced38`); configuration `configs/disclosure_targeted_audits_confirmatory_v1.json`, experiment ID `06bec148...2a32`, seeds 42001–42030; report `reports/disclosure_targeted_audits_20261020/`.
- *Assurance providers:* research contract, Section 22, committed before implementation and runs (`307168c`); pipeline checks, frozen configuration and analysis script committed before the confirmatory run (`61b1205`); configuration `configs/disclosure_assurance_confirmatory_v1.json`, experiment ID `35c4f767...339a`, seeds 43001–43030; report `reports/disclosure_assurance_20261021/`.
- *Learning-firm redo:* research contract, Section 23, committed before implementation and runs (`8bb5d9e`); frozen configuration committed before the confirmatory run (`42ec591`); configuration `configs/disclosure_learning_redo_confirmatory_v1.json`, experiment ID `b4430136...6062`, new seed blocks listed in contract Section 23.1; report `reports/disclosure_learning_redo_20261021/`.
- *Joint robustness test:* research contract, Section 24, committed before implementation and runs (`76308ea`); pipeline checks, frozen configuration and analysis script committed before the confirmatory run (`be0f850`); configuration `configs/disclosure_joint_robustness_v1.json`, experiment ID `c856a0cd...6765`, seeds 44001–44030; report `reports/disclosure_joint_robustness_20261023/`.
- *Figures:* `pythonpaper/make_figures.py` reads only committed analysis files.
- *Browser version:* the firm decision rule in the online sandbox matches the Python rule on 1,200 random inputs with a maximum relative error of 1.0×10^{-14}.

C Additional results

Table 11: Dose responses (true change relative to B, SD; share of decisions that inflate)

Transparency τ	0.3	0.5	0.6	0.7	0.8
Change	+0.181	+0.259	+0.331	+0.337	+0.340
Inflating	11.3%	8.0%	0	0	0
Fixed fine (multiple)	0×	1× (B)	2×	4× (F)	8×
Change	−0.414	0	+0.088	+0.125	+0.143
Inflating	90.6%	22.3%	15.5%	13.3%	12.5%
Proportional fine (multiple)	1×	2×	4×		
Change vs. same fixed multiple	−0.338	−0.358	−0.290		
Inflating (large / mid / small)	72 / 80 / 88%	62 / 72 / 81%	48 / 57 / 66%		